Ex. 6: Analog Beamforming

Code

```
 clc;
clear;
close all;
lambda = 1;
d = lambda/2;
theta = -90:0.1:90;
theta_rad = deg2rad(theta);
N_array = [8 16 32 64];
beamwidth = zeros(size(N_array));
for idx = 1:length(N_array)
N = N_array(idx);
theta0 = 0;
theta0_rad = deg2rad(theta0);
AF = zeros(size(theta_rad));
for n = 0:N-1
AF = AF + exp(1j*2*pi*(d/lambda)*n.*...
(sin(theta_rad)-sin(theta0_rad)));
end
AF = abs(AF);
AF = AF/max(AF);
[~, peak] = max(AF);
left = peak;
while left > 1 && AF(left) >= 1/sqrt(2)
left = left - 1;
end
right = peak;
while right < length(AF) && AF(right) >= 1/sqrt(2)
right = right + 1;
end
beamwidth(idx) = theta(right) - theta(left);
end
disp('===== 3-dB Beamwidth Results =====')
for i = 1:length(N_array)
fprintf('N = %2d --> Beamwidth = %.2f degrees\n', ...
N_array(i), beamwidth(i));
end
figure;
plot(N_array, beamwidth,'-o','LineWidth',2,'MarkerSize',8);
grid on;
xlabel('Number of Antenna Elements (N)');
ylabel('3-dB Beamwidth (degrees)');
title('Beamwidth vs Number of Antenna Elements');
```

Experiment7.m

/MATLAB Drive/5g6g/Experiment7.m

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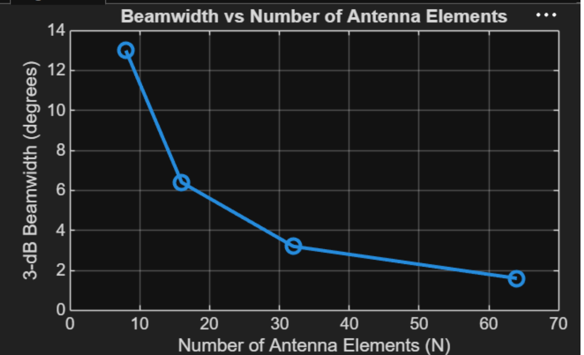
theta_rad = deg2rad(theta);
N_array = [8 16 32 64];
beamwidth = zeros(size(N_array));
for idx = 1:length(N_array)
    N = N_array(idx);
    theta0 = 0;
    theta0_rad = deg2rad(theta0);
    AF = zeros(size(theta_rad));
    for n = 0:N-1
        AF = AF + exp(1j*2*pi*(d/lambda)*n.*...
            (sin(theta_rad)-sin(theta0_rad)));
    end
    AF = abs(AF);
    AF = AF/max(AF);
    [~, peak] = max(AF);
    left = peak;
```

Figure 1



Command Window

```
N = 16 --> Beamwidth = 6.40 degrees
N = 32 --> Beamwidth = 3.20 degrees
N = 64 --> Beamwidth = 1.60 degrees
>>
```